



The effect of light on the growth rate of intertidal *Acropora pulchra* (Brook) from Phuket, Thailand, lat. 8°N

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Accepted 13 January 1987

Abstract. Series of apices of *Acropora pulchra* from an intertidal reef at Phuket, Thailand, were grown at different depths in the sea, and the length growth was monitored at 12–24 h intervals with laser diffraction. The growth rates of actively growing apices were normally distributed and showed a high variability, with an average coefficient of variation of 58%. There was a highly significant difference in average growth rate between adjacent colonies. A significant linear relationship was found between irradiance and length growth, with a saturating level at 300–400 Wm⁻². At 1000 Wm⁻² length growth was significantly reduced. Under normal daylight conditions, day and night growth rates were equal.

Introduction

Light has been shown to affect length growth and calcification in corals (e.g. Kawaguti and Sakumoto 1948; Goreau 1959, 1963). This may be related to light dependent behaviour during feeding, to calcium carbonate deposition processes, to the activity of the zooxanthellae or to other light influenced processes related to growth and calcification. It has been generally accepted that the effect of light is more or less directly related to photosynthesis (see Crossland and Barnes 1977; Vandermeulen and Muscatine 1974). This view has recently been disputed by Rinkevich and Loya (1984), who suggest that the immediate stimulation of calcification by light is not directly correlated with algal photosynthesis. Attempts to correlate skeletal deposition with light intensity or with day and night cycles, show conflicting results (Goreau and Goreau 1959; Chalker and Taylor 1975; Strömberg 1976; Gladfelter et al. 1978; Barnes and Crossland 1980; Oliver et al. 1983; Rinkevich and Loya 1984).

Although some observations indicate that total coral calcification and length growth of apices to some degree are not completely phased (Barnes and Crossland 1980), growth rates of apices are closely linked to apical calcifi-

cation. In the present experiment a highly sensitive and non-destructive method, laser diffraction (Strömberg 1975), is used to investigate the short term (5–7 days) growth rates of apical tips of intertidal *Acropora pulchra* measured during 12–24 h intervals in a depth/irradiance gradient, and during day/night cycles.

Materials and Methods

In the Phuket area, *A. pulchra* grows in the intertidal zone, where it is periodically exposed to direct sunlight for up to 40% of the tidal cycle and where the water depth seldom exceeds 1–1.5 m. Coral tips of length 1.5–2.5 cm were sampled at low tide from several colonies within a narrow and uniform area of the reef. The tips were strapped rigidly to plastic frames with rubber bands, and immediately submerged in the sea. The experimental site was close to the coral reef (<50 m), with approximately identical environmental conditions.

The elongation of the tips was measured with laser diffraction (Strömberg 1975). During the measurements, the corals were exposed to air for 3 min and to weak NeHe laser light (632.8 nm, 1 mW) for <15 s.

Surface light was monitored with a lux-meter with a flat sensor at 30 min intervals. The lux-meter was later calibrated in sunlight against a Meteorologic radiometer with a flat sensor, and the lux values converted to Wm⁻² (conversion factor: 1 Wm⁻² ~ 100 lux). Attenuation profiles were based on underwater light measurements with an in situ quantum light meter. Due to tidal currents, the water convection is good at the experimental site, and the temperature and salinity is uniform in the water column, 28.0–28.2 °C and 32.0–32.1‰.

Three different experiments were carried out:

1. Fifteen colonies of *A. pulchra* were sampled within a 10 m² area. From each colony were subsampled 20 white tips of length 15–25 mm. The mounted tips were placed horizontally in the sea, 0.5 m below the sea surface. After 24 h of acclimatization, the elongation of the individual tips was measured during 24 h intervals for 5 days.

2. Four series of 50 tips each, subsampled at random from a large batch ($n \sim 600$) of white tips from several colonies growing within a 2 m² area of the reef, were placed in open cages at 0.5, 1.0, 2.5, and 5.5 m below the sea surface. One series of 60 individuals was placed at 0.5 m of depth in cages covered with black polyethylene sheet on the top, and with loose flaps hanging along the sides. This arrangement gives minimal resistance to the horizontal flow of water across the coral apices. Finally, one series of 60 individuals was submerged 1–2 cm below the water surface in channel-formed aquaria (6 l), and supplied with a rapid flow of unfiltered seawater, pumped from 0–2 m of depth (1 l min⁻¹). All mounted tips were acclimatized at 0.5 m of depth for 48 h. The growth was measured during 24 h intervals for 7 days.

3. One series of 60 individuals was placed in cages at 0.5 m of depth. The tips were sampled and acclimatized as in experiment 2. The length growth was measured every 12 h, at sunrise and sunset, for 7 days. This experiment was repeated with new apices for 4 days. A parallel series was measured at the start and the end of each experiment.

Results

The growth rates of all white tips ($n=300$) used in experiment 1 were normally distributed (Fig. 1), as were also the growth rates within each colony ($n=20$) (Kolmogorov-Smirnov test for goodness of fit). Average growth rates of the 15 separate colonies grown at 0.5 m of depth and measured in 24 h intervals for 5 days, ranges from $0.6 \mu\text{m h}^{-1}$ to $4.6 \mu\text{m h}^{-1}$, with standard errors of mean ranging from 0.12 to 0.58, respectively (Table 1). The average coefficient of variation (CV) was 58%, with maximum range 35% to $>100\%$, dependent on time scale. Among the different colonies, grown under identical conditions both before and during the experiment, there was a highly significant difference in average growth rate. Experiment 1 was carried out during a period with clear sky, and with maximum noon irradiance of $\sim 1000 \text{ Wm}^{-2}$ at the sea surface. The irradiance at 0.5 m of depth was 85% of that at the surface.

During experiments 2 and 3, the irradiance in 0.5, 1.0, 2.5, and 5.5 m of depth was 90%, 65%, 40%, and 15%

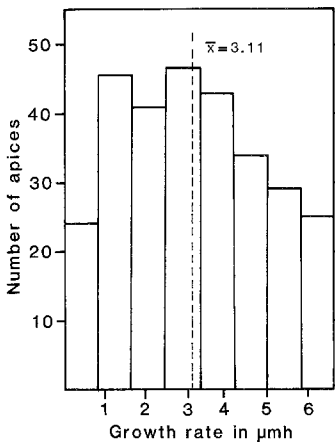


Fig. 1. Distribution and growth rates of actively growing apices of *Acropora pulchra* ($n=300$)

Table 1. Average growth rates of actively growing apices from 15 adjacent colonies of *Acropora pulchra*. Number of apices per colony is 20. SE indicated

$\bar{x}$	SE	$\bar{x}$	SE
0.69	0.12	3.49	0.50
1.22	0.23	3.53	0.50
1.78	0.40	3.73	0.32
1.87	0.29	3.95	0.48
2.12	0.30	4.24	0.53
2.46	0.21	4.30	0.49
3.20	0.37	4.58	0.58
3.42	0.54		

of surface light, respectively. In the cage covered with black polyethylene (experiment 2), the light was only 1% of that at the surface. During the experiments, the sky was lightly and uniformly clouded with maximum noon surface irradiance about 600 Wm^{-2} , and this indicates maximum noon irradiances of 540, 390, 240, and 90 Wm^{-2} at 0.5, 1.0, 2.5, and 5.5 m of depth, respectively. The corresponding irradiances in the cage covered with black polyethylene was 6 Wm^{-2} . At the 7th day of the first day/night growth experiment and at the experiment where the apices were grown in aquaria, the sky was clear with maximum surface irradiance of about 1000 Wm^{-2} .

The effect of frequent measurements on the growth rate of *A. pulchra* was estimated by comparing apices grown in aquaria at 1–2 cm of depth and measured every 12 h for 6 days, with a parallel series measured only in the beginning and the end of the period. Those measured frequently had an average growth rate of $2.16 \mu\text{m h}^{-1}$ ($n=60$, $\text{SE}=0.18$) compared to $1.96 \mu\text{m h}^{-1}$ ($n=60$, $\text{SE}=0.13$) for the control. This indicates that the measuring procedure does not interfere significantly with normal growth during short term experiments. The laser light used during the measurements has an irradiance that is approximately that of zenith tropical sun, and no harmful effects on growth was apparent with continuous exposure up to 10 min.

When the apices were grown at different depths for 7 days, no significant differences in growth rate were found between 0.5 (540 Wm^{-2}) and 1.0 m (390 Wm^{-2}) of depth. The average growth rate in these depths was 4.38 and $4.18 \mu\text{m h}^{-1}$, respectively (Fig. 2). At 5.5 m (90 Wm^{-2}) a significant decrease occurred within 3 days, and in the dark cages at 0.5 m (6 Wm^{-2}) within the first 48 h, reaching a minimum level within 4–6 days. When the growth rate after 4 and 6 days of exposure is plotted against irradiance, a linear relationship is indicated up to

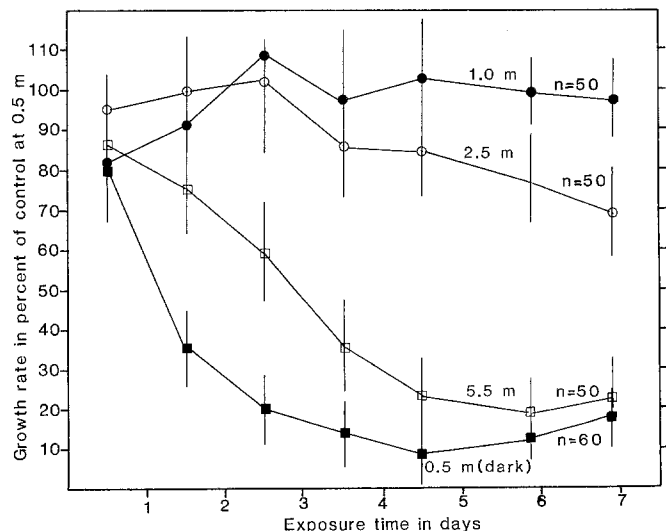


Fig. 2. Growth rates of *Acropora pulchra* at different depths in the sea in percent of growth rate in 0.5 m. SE indicated

300–400 Wm^{-2} (Fig. 3). The growth rate of tips grown in nearly darkness at 0.5 m of depth is about 12% of that of tips grown at the same depth in full sunlight. This corresponds to a dark to light ratio for apical growth of about 1:8.

The tips grown in aquaria 1–2 cm below water surface and exposed to maximum irradiances of nearly 1000 Wm^{-2} became bleached on the exposed side, but a fairly rapid growth still prevailed in the end of a 6 days period of about 60% of the initial growth rate.

Day (D) and night (N) growth rates were equal (Fig. 4) during the two separate experimental sequences, with grand average $D=2.31$, $N=2.65$ and $D=2.14$, $N=2.20$, respectively. The high growth rates the last day and night of the 7 days sequence are related to an increase in maximum noon irradiance at 0.5 m from 540 Wm^{-2} to about 900 Wm^{-2} , and are not included in the grand average.

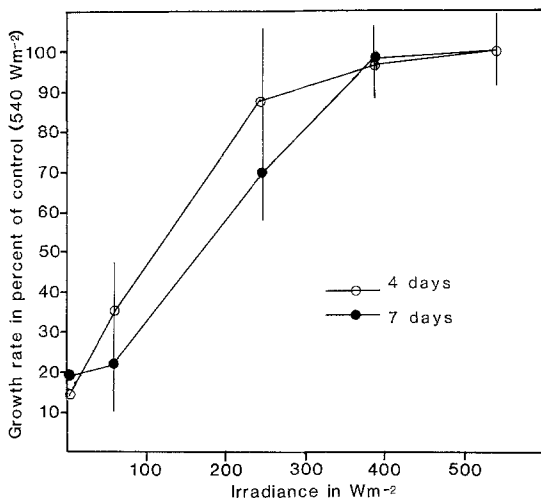


Fig. 3. Growth rates in percent of growth rate at 540 Wm^{-2} of *Acropora pulchra* exposed to different irradiances for 4 and 7 days. SE indicated

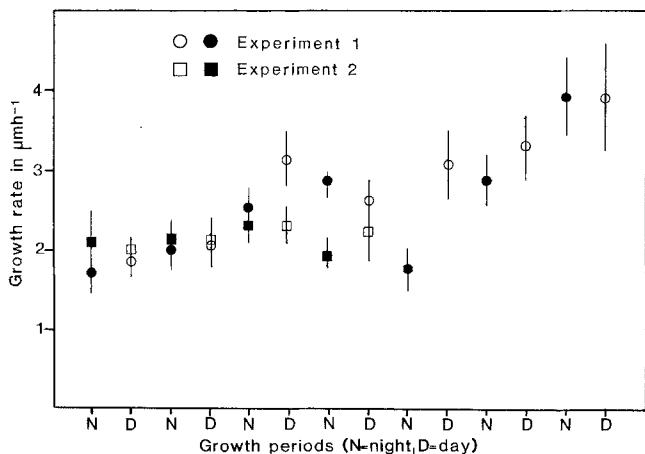


Fig. 4. Growth rates ($\mu\text{m h}^{-1}$) of *Acropora pulchra* ($n=60$) during day (0600–1800 hours, open symbols) and night (1800–0600 hours, filled symbols). SE indicated

Discussion

The tips of the upper and central part of the *A. pulchra* colonies were more or less brownish due to zooxanthellae pigment, while the lateral tips were usually white, indicating fewer zooxanthellae. Preliminary experiments showed that the brownish or tinted tips had no growth or grew only slightly compared to the white tips. Oliver et al. (1983) refer to similar observations for *A. formosa*, and Goreau (1959) observed that large apical polyps of some of the branching acroporids contained few zooxanthellae, but calcified several times faster per unit tissue nitrogen than the yellowish brown lateral polyps which were literally stuffed with algae.

A consequence of the growth pattern observed in *A. pulchra*, with two main groups of apices with widely different growth rate within a single colony, is that overall colony averages are of restricted value. Furthermore, the proportion white tips/coloured tips varied widely among the different colonies, and if growth of different colonies are compared, based on a random sampling of apices from each colony, the results may be misleading. If the purpose of an investigation is to estimate total average apical length growth of a colony or a population, both types of apices should be considered according to their relative abundance, but if relationships are sought between apical growth and environment, only white tips may be used.

Large variance is typical for coral growth. Barnes and Crossland (1982) stated from a review of coral literature that coral calcification was a highly variable process over time, within and between coral colonies, and within any group of experimental replicates, and Goreau and Goreau (1959) and Barnes (1982) state that apparently similar pieces of coral from the same corallum can have very different calcification rates.

Although all apices were sampled from a uniform part of the reef and grown under identical conditions, the variability in growth rate of the actively growing white tips is large, both within and between groups (Fig. 1, Table 1). Polyps of a colony are clones with genetically the same growth potential, and Isdale (1977) suggested that a wide variation in growth rate of adjacent specimens of *Porites lobata* might be related to patchy distribution of food, or to genetic differences.

The average coefficient of variation (CV) of growth rate for *A. pulchra* from Phuket, based on all experiments, was 58%. For the same species, Yap and Gomez (1981) present data which show that the average CV for monthly length growth measured throughout the year was 80%. This high variability implies that large samples are required to detect differences between growth rates of two populations. With a CV of about 60%, about 70 replicates are required to detect a 20% difference between two means at the 0.05 level.

The growth rates of *A. pulchra* from Phuket are low (average 3.11 to 4.38 $\mu\text{m h}^{-1}$ at 0.5 m of depth), compared to growth rates of the same species from the Phil-

ippines of $27 \mu\text{m h}^{-1}$ during December–February (Yap and Gomez 1981). The specimens from the Philippines, however, are subtidal, while the Phuket reef is intertidal and regularly exposed to desiccation, combined with strong solar radiation and high water temperature. During low tide, feeding cannot take place and the environment probably represents conditions close to the physical tolerance limits for the coral. Furthermore, the growth of cut apices of *Acropora* may be different from intact ones due to reduced translocation of photosynthate. Pearse and Muscatine (1971), Taylor (1977), and Oliver et al. (1983) observed that excised tips of *Acropora* calcified slower and did not show light enhancement of calcification. In the present experiments, however, length growth was considerable and stable, and responded significantly to changes in light conditions, and it is reasonable to assume that excised tips may be used for short-term investigations on relationships between apical growth and environmental parameters.

Length growth in corals is closely related to calcification, which in turn is dependent on several processes; uptake and assimilation of food, transport of assimilated material, production of an organic matrix, and deposition of calcium carbonate (Goreau 1959; Barnes 1971). All these processes may be influenced by light intensity or by the relative duration of light and darkness. Feeding in corals is partly autotrophic by zooxanthellae, partly heterotrophic on a wide variety of food: bacteria, detritus, dissolved organic matter, and animal prey (Muscatine 1973; Porter 1974; Sorokin 1982). Keeping the corals at different depths in the sea may thus affect the irradiance, the spectral distribution of the light, and the composition of the food available. It is assumed, however, that irradiance is the most relevant component of the light, and that the rapid convection of water at the experiment site tends to counteract a vertical food gradient along the cages.

The close relationship found between length growth rate and depth/irradiance (Figs. 2 and 3) is in accordance with previous observations, indicating that length growth and calcification in corals are closely related to the activity of the zooxanthellae (e.g. Kawaguti and Sakumoto 1948; Goreau 1959, 1963; Pearse and Muscatine 1971; Vandermeulen et al. 1972; Chalker and Taylor 1975; Schmitz and Kremer 1977; Leletkin and Zvalinsky 1982). The growth/irradiance relationship is linear up to $300\text{--}400 \text{ Wm}^{-2}$ (Fig. 3). This corresponds to the observation by Chalker and Taylor (1975), that calcification in *A. cervicornis* was directly proportional to subsaturating light intensities between 0 and 10 500 lux, corresponding to about 0 to 100 Wm^{-2} . However, Gladfelter et al. (1978) and Oliver et al. (1983) report that extension rates of *Acropora* increased with depth, and Oliver et al. (1983) suggested this to be an adaptation to sub-optimal conditions by increasing the chance of growing into more favourable environment. Rinkevich and Loya (1984) propose that the immediate stimulation of calcification by light is not directly correlated with algal photosynthesis

and that the significant decrease in calcification rate in dark control experiments could be the result of oxygen depletion of the water. Oxygen limitation did certainly not occur in the present investigation.

The saturation of the light dependent growth at $300\text{--}400 \text{ Wm}^{-2}$ may represent a functional saturation level for photosynthesis of the zooxanthellae involved, and corresponds to photosaturation in oxygen exchange rates in *A. acuminata* at about 300 Wm^{-2} , observed by Crossland and Barnes (1977). This is considerably higher than the light saturation for gross photosynthesis by 3-cm tips of *A. cervicornis* at about 100 Wm^{-2} , reported by Chalker and Taylor (1978). In very strong sunlight (1000 Wm^{-2}), photoinhibition of growth occurred, combined with bleaching of the zooxanthellae on the exposed side of the coral. In this case the corals were kept only 1–2 cm below the water surface, and strong UV-radiation is thus involved. Photoinhibition of calcification at surface light intensities (abt 400 Wm^{-2}) are indicated in shallow corals (Barnes and Taylor 1973; Baker and Weber 1975; Maragos 1972), while Muscatine (1980) concludes that photoinhibition has not been demonstrated in corals.

In continuous low light for more than 24 h (noon irradiance 6 Wm^{-2}), the growth of *A. pulchra* is further reduced, and after 4–6 days of exposure, a minimum level is reached with the growth rate being only 12% of that in normal daylight at the same depth (0.5 m). This gives a L/D ratio of 1:8 (Fig. 2). In *Millepora complanata* Strömberg (1976) observed a reduction of growth rate to only 4% in continuous darkness compared to normal daylight. These observations suggest that autotrophic feeding is decisive for rapid growth.

When *A. pulchra* was grown in normal daylight at 0.5 m of depth, the growth rate during night was equal to the day growth rate (Fig. 4). Barnes and Crossland (1980) using a photographic technique also observed equal day and night growth in *A. acuminata*. For *Millepora complanata* Strömberg (1976) found that night growth rates were slightly larger than day growth rates, and Roth et al. (1982) interpreted this observation as a delay from ingestion/assimilation to growth. Pearse and Muscatine (1971) found short-term light/darkness (L/D) calcification ratios of *A. cervicornis* of about 2 to 3 for intact tips and 1.5 for isolated tips. Sorokin (1982) states that dark to light ratios of coral calcification vary from 1:2 (50%) to 1:9 (11%), and are usually around 1:2–3. In *M. complanata* Goreau and Goreau (1959) found that the calcification rate in darkness was reduced to 10–15% of that in daylight. The large difference in L/D ratio between 24 h and 4–6 days of exposure, indicate that the L/D ratio is very dependent on exposure time.

High night growth rates in *Acropora* and *Millepora* may depend on zooxanthellae activity during the previous day hours, and this indicates that the pool of photosynthates produced during the day are distributed more or less evenly throughout the 24 h growth period. In contrast to most other corals, the polyps of *Acropora* spp.

and *Millepora* spp. are fully expanded both during day and night (Porter 1974). This may suggest that there is no diurnal variation in ability to catch particulate food or dissolved organic matter. However, Sorokin (1982) found that the ingestion by corals of *Artemia* nauplii was 5–15 times greater during night than during day, and that the highest rates of predatory feeding occurred in corals with small polyps, e.g. *Acropora*. This feeding habit may compensate for reduced supply of photosynthate during dark hours. Barnes (1970, 1972) demonstrated the existence of daily patterns in the deposited skeleton of hermatypic corals, indicating a diurnal difference in activity. An equal length growth during day and night should not produce daily patterns of growth bands in *Acropora* and *Millepora*, unless light and darkness influences the structure of the calcium carbonate deposited.

Acknowledgements. Thanks are due to the staff of Phuket Marine Biological Center for their assistance. Dr. P. Boonruang kindly provided the underwater light measurement data.

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